

Automated Smart Attendance System for Higher Education using a Lightweight DNN Model-based Face Recognition

1. Shashank Shekhar Tiwari, Assistant Professor, Department of Information Technology, Rajkiya Engineering College, Dr. Abdul Kalam Technical University, Lucknow, U.P, India, er.sst.010@recabn.ac.in

2. Dr. Sushil Kumar, Department of Education, Bhagat Phool Singh Mahila Vishwavidyalaya, Khanpur Kalan, Sonapat, Haryana, India, drkumar230123@gmail.com

3. Dr. RVS Praveen, Digital Engineering and Assurance, Skyview Building No 20, Sarelingampally Hitech City, Hyderabad, Telangana, India, praveen.rvs@gmail.com

4. Gautam Kumar, NIELIT Itanagar, Arunachal Pradesh, India, gautamkumarconference@gmail.com

5. Savanam Chandra Sekhar, Department of BBA, KL Business School, Koneru Lakshmaiah Education Foundation, KL University, Guntur, Andhra Pradesh, India, savanam.sekhar@gmail.com

6. Rajashree Bhokare, Department of Electrical Engineering, Dr.D.Y.Patil Institute of Technology Pimpri, Pune, Maharashtra, India, rajashree.bhokare2020@gmail.com

Abstract – This research introduces a novel online system for automated student attendance utilising modern technologies such as Streamlit, Redis, Python, and facial recognition. Conventional attendance methods, like human roll calls and barcode scanning, are susceptible to inaccuracies, deceit, and inefficiency. The suggested system utilises deep learning methodologies to develop a resilient facial recognition model, offering a precise and efficient alternative. Feature extraction is performed via the Local Binary Patterns Histogram (LBPH) technique, and the model is executed with the DS-CNN-LSTM architecture to improve efficacy. Validation on the Smart Attendance dataset confirms the system's efficacy, attaining a maximum training recognition rate of 97.56% and a validation accuracy of 95.32%. Cross-validation and temporal complexity analyses substantiate the model's dependability and computational efficacy. This method provides a reliable, efficient, and fraud-resistant alternative for attendance management in educational environments.

Keywords— Smart Attendance System (SMS), Face Recognition (FR), Facial Detection, Depthwise Separable Convolution Neural Network Long Short-term Memory Networks (DS-CNN-LSTM), Fully Connected Layer (FC).

I. INTRODUCTION

Managing student attendance in higher education is challenging, particularly in large classrooms, as manual techniques for recording attendance require significant time and effort for lecturers to recognise or recall students' faces. Furthermore, digitising the time attendance system poses challenges as it is crucial for expediting the decision-making process and reducing the likelihood of failure. In Higher Educational institutions requiring comprehensive internet connection throughout their premises must transition to an automated attendance system. Managing employees is one of the most challenging problems in organisations, and maintaining traditional paper attendance records or sheets adds to the complexity of the manual or traditional time attendance system. Numerous educational institutions employ non-academic personnel to record student attendance daily,

which is time-consuming rather than resource-efficient, as it necessitates periodic updates of materials such as paper and storage for these data [1]. Attendance tracking is a routine practice in organisations and enterprises.

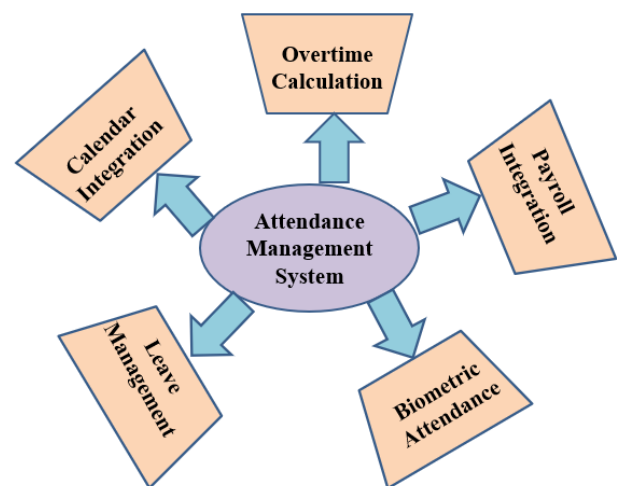


Fig. 1. Attendance Management System Applications

Software and production companies are merely two examples of the several industries utilising this initiative to guarantee staff attendance at all times (Figure 1) [2]. An attendance system is essential for any organisation that wants to keep track of when students and employees are present. They possess their unique methodology to get the same outcome. Some individuals record attendance manually, while others utilise an automated attendance system. The manual technique involves the use of pen and paper, resulting in significant time consumption and resource wasting. Additionally, it poses risks associated with proxies and human error. Face recognition is one of the most effective technologies available for the identification of individuals. It is applicable in schools, colleges, or any organisation [3]. To mitigate the challenges associated with recording attendance for a substantial number of individuals, an automated attendance system is necessary that is efficient

and minimises the likelihood of fraudulent attendance. This technology solution is designed to facilitate a simple and secure method for recording attendance. Attendance is documented by continuously monitoring the faces of students through a webcam as they enter the classroom. The software simultaneously checks the faces with the pre-established database after first detecting them.

This Paper is structured as follows. The subsequent section delineates the pertinent literature. Section 2 addresses the problem statement. Section 3 presents the suggested system architecture. Section 4 encapsulates the implementation, whereas section 5 closes the paper.

II. LITERATURE SURVEY

The primary aim of this work is to create an intelligent attendance management system utilising facial recognition to address the issues encountered in existing automated systems already in use. [4] examined several face recognition systems, including PCA, Eigenface, Support Vector Machines (SVM), and Neural Networks, and compared them based on their efficacy rates. The authors also discussed system architecture, a step-by-step technique, and substantiated it using its algorithm. They have also presented a mathematical model employing mathematical notions and terminology. The model introduced by [5] utilises cutting-edge developments, employing CCNN for face identification and CNN for face embeddings. CNNs yield optimal results with larger datasets; conversely, the initial evaluation involved implementing these techniques on smaller datasets. The overall precision was 95.02% based on a small dataset of first facial images of workers in the current situation. This paper [6] proposes a concept for an automatic attendance system utilising face detection. This system primarily concentrates on algorithms for face detection and recognition. This system operates with high efficiency. It can readily identify the students upon their entrance into the classroom. If it corresponds with any of the database photographs, he/she will be recorded as present for the specific session; otherwise, he/she would be tagged as absent. This system primarily comprises preprocessing, feature extraction, image capture, database construction, classification, and post-processing stages. These phases constitute the primary components of the automatic attendance marking system. [7] utilised facial recognition for classroom attendance. The paper utilised Eigenface for recognition; however, it did not specify the total accuracy of the system. A similar method was suggested in [8] for the face recognition-based attendance tracking system utilising Eigenface. They were able to identify faces with an overall identification accuracy of 85%. [9] employed Principal Component Analysis (PCA) for facial recognition in their attendance system, attaining an accuracy of 68%. The swift advancement in deep learning has significantly enhanced the accuracy of face recognition through the application of deep CNNs. [10] introduced the innovative system – FaceNet, which relies on Deep Neural Networks (DNN) for face identification. The suggested method attained remarkable results on the Labelled Faces in the Wild (LFW) dataset, with 99.63% accuracy.

In order to automatically take attendance, [11] suggested a system that uses facial recognition technology to identify individual pupils. The algorithm attained an accuracy of 93.1% in student face recognition. Conversely, noise in images considerably affects recognition performance, leading to a decline in total accuracy. The precision attained by the suggested attendance methods is inadequate for dependable real-time student identification. Consequently, the significant concern regarding recognition accuracy must be addressed. This research proposes a solution to address the aforementioned issues.

This paper's primary contributions to the detection of epileptic seizures from EEG signals are outlined as follows. The output of DSCNN serves as the input for training the LSTM model, which addresses the issues of gradient vanishing and gradient explosion during long sequence training. Furthermore, the temporal attributes can be derived using LSTMs. These elements are primarily essential for modelling calculations, while also indirectly assisting in the attendance management system.

III. PROPOSED SYSTEM

When the class size is large, it could be challenging to keep track of attendance while still actively engaging the kids. Taking attendance by hand is a time-consuming and laborious process, and students can always be excused [12]. It designed a smart attendance management system (SAMS) that uses location, fingerprints, and facial recognition to solve this challenge and keep track of students' attendance.

In contemporary society, the world is undergoing rapid digitalization, resulting in a significant transformation in the assessment of life. Individuals are increasingly aware of time consumption and seek to implement new technologies to facilitate their tasks. The Smart Attendance System enables the monitoring of student presence in the classroom and provides updates to notify their guardians [13]. The system has been trained with six distinct students, utilizing over 100 images per student, which were converted to grayscale and stored alongside the student's name and ID.

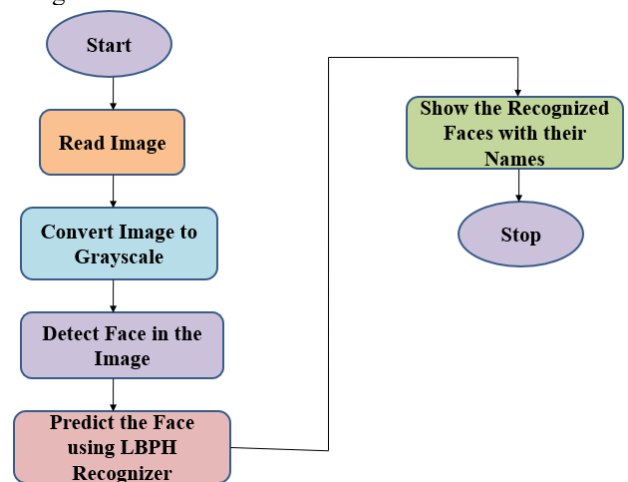


Fig. 2. Flow Chart of the Methodology

The training procedure commences with the navigation of the training data directory. As show in Fig. 2 each image in the training dataset is transformed to

grayscale. A section of the image is designated as the centre, and its neighbouring elements are evaluated against it using a threshold. Separate lists for each face have been created, and the faces, along with their corresponding IDs, have been appended to these lists. The faces are subsequently taught using their corresponding IDs.

A. Recognition of Face:

The recognition technique revolves around face recognition. One kind of computer vision is face recognition, which uses data about a person's facial features to determine their identification. To put it simply, there are two main components to face recognition: detection and matching. In order to identify a person, face recognition software analyses their unique facial characteristics in conjunction with an input video or photograph of their face [14]. Find out if it's a human face first, and if so, specify where it is, how big it is, and where each of the main facial organs is located. Using this data, we may extract the face's identification traits and compare them to those of known faces to determine the face's identity. The four main components of face recognition technology—which is a subset of biometric recognition technology—are as follows: hardware cameras, network lines, computing devices, and feature extraction from facial images. The final step is matching and combining the extracted features with the collected images. Here is the method of calculation in Eq. 1 :

$$S = \min\{S_1, S_2, \dots, S_m\} \quad (1)$$

Facial recognition theory. One sort of biometrics technology is facial recognition, which uses pre-processing and data acquired from cameras to identify individuals. Recognition of faces automatically. Gathering data to identify the presence or absence of a human face in an image, as well as the image's size and location, and then segmenting the identified human face into an adult facial region, are the primary goals of face detection. Finally, there's face recognition, which uses extracted facial feature information and image data to check if the data is in the repository. There won't be any recognition outcomes unless it is, as it has matching identification information.

B. Feature Extraction and Matching:

Matching the ROI face data from the prior face detection procedure was the first step in this procedure. In order to compensate for changes in lighting, the alignment procedure included resizing the face. Because of the identified face size, the varying distances between the face and the camera, and the differences in environmental light intensity during image capture, this technique was essential to ensure that the next step would produce consistent results. The LBPH method was used to extract the features [15]. Finding the most comparable face in the database would be the goal after detecting a face. For the sake of clarity, the database's names and directories are displayed in the CSV file format. The blue circle represents the user's eye position, which the feature extraction findings were able to detect. In subsequent steps, the user's nose and mouth positions would be used in conjunction with the

distance between the circle's centre to carry out the identification and matching process.

C. Model Training:

This work proposes a CNN-LSTM model with one input layer, one depth-separable convolution layer, one pooling layer, two full connection layers, two LSTM layers, and one output layer. The model is one dimensional. The addition of a drop layer helps to avoid overfitting. Figure 4 shows the schematic of the model. The usage of few neurons in the proposed Depthwise Separable CNN-LSTM model is a benefit of depthwise

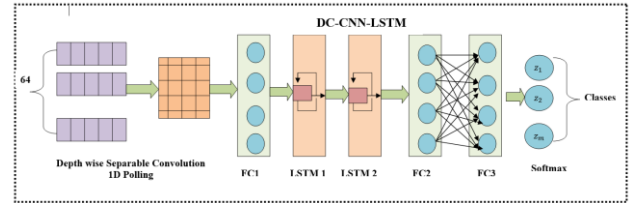


Fig. 3. Flow Chart of the Methodology

Figure 3. Structure of the Depthwise Separable Convolutional Neural Network and Two-Layer Long Short-Term Memory Networks (DS-CNN-LSTMs) model. FC represents the fully connected layer, while y denotes the classification category.

The first step is to feed the 1d EEG data that has already been processed into the model's input layer; this data has 178×1 dimensions. Then, in order to extract characteristics of EEG signals, the input data is subjected to a one-dimensional depth-separable convolution procedure. Here is the exact operation that is being performed: The first separable Conv1D layer has 64 one-dimensional convolution kernels with a 3×1 size and a 1 step size. The convolution kernels stand in for the convolutional sensory field. Inadequate sensory fields and narrow convolution kernels prevent it from efficiently extracting association characteristics between nearby characters across a wider range. When the convolution kernel is either too tiny or too big, it becomes easier to overlook the association features between nearby adjacent characters, which in turn leads to inaccurate classification results. The ReLU nonlinear rectification linear unit and a 31×1 convolution kernel size are suitable for many tests. To prevent the over-fitting issue, the ReLU activation function is used. In Equation (2), we can see the ReLU equation:

$$\pi(w) = \begin{cases} 0, & w \leq 0 \\ w, & w > 0 \end{cases} \quad (2)$$

Next, it moves on to the pooling layer, which keeps the most important features while decreasing computation and parameters (latitude) to avoid over-fitting. This layer follows the one-dimensional convolution layer. After passing through the pooling layer, the fully connected layer incorporates a dropout layer to curb overfitting [16]. The output features are supplied into the LSTMs layer after passing through FC Layer1. This layer has the capability to learn valuable information from EEG time series data. In both the first and second layers of LSTMs, there are 64 neurone units. The output characteristics are transferred to another FC Layer2 after passing through the LSTMs layer.

The 32-neurons FC Layer2 stores the final model data as features, feeds those features into the softmax layer for classification, and then outputs the model. To make sure the probability isn't negative, the softmax classifier first transforms the model's prediction results into an exponential function. Normalisation must transform the result to ensure that all of the predictions have probability equal to 1. You can think of it as a percentage of the entire number of converted outcomes; the process is dividing the converted result by the sum of all the converted results. I can give you a rough idea of the likelihood then. By doing so, the last feature vector will be transformed into a set of values between 0 and 1, which, when added together, form a probability of 1. At the end of the prediction process, the output node will be the one with the highest probability. Eq. (3) shows the value of softmax.

$$O(z|w) = \frac{f^{g(w,z_j)}}{\sum_{i=1}^m f^{g(w,z_i)}} \quad (3)$$

This experiment employs categorical cross entropy loss and the Adam optimizer, utilising cross entropy to assess the disparity between the probability distribution derived from training and the actual distribution. It delineates the disparity between the actual and anticipated output probabilities; specifically, a lower cross-entropy value indicates a greater proximity between the two probability distributions. The Adam optimizer integrates the benefits of AdaGrad and RMSProp, two distinct optimisation methods. The update step size is determined by evaluating the First Moment Estimation and Second Moment Estimation. Adam is selected as the optimizer because to its simplicity and computational efficiency as a stochastic gradient descent method.

$$\tau = \pi(y), \text{ where } y = \sum xi \times wi + a \quad (4)$$

$$D = -\frac{1}{m} \sum_w [zkmb + (1 - z)km(1 - \tau)] \quad (5)$$

where z represents the anticipated output; b denotes the actual output of the neurone.

$$\frac{\partial d}{\partial x_i} = \frac{1}{m} \sum_w w_i (\pi(y) - z) \quad (6)$$

$$\frac{\partial d}{\partial a} = \frac{1}{m} \sum_w (\pi(y) - z) \quad (7)$$

The empirical findings indicate that Adam outperforms other stochastic optimisation techniques. The model's setup can be tailored to the specific requirements of the identification task [17]. The formula for the cross-entropy loss function is presented in Equations (4)– (7), including the derivative transformation.

Algorithm Attendance System for Face Recognition

Input: Image of the classroom obtained by the camera.

Output: Marking Attendance.

Problem Description: Student identification

Step 1: Start

Step 2: Register the students' details in the facial recognition database.

Step 3: Install a surveillance camera in the classroom.

Step 4: Upload the image captured by the camera.

Step 5: Image enhancement.

- Transform into a grayscale image.
- Produce a histogram of the grayscale image.
- Normalise the image.
- Create a histogram of the equalised image.
- Eliminate noise from the image
- Classification of skin in images.

Step 6: Facial Detection.

- Extract the faces of students from the image.
- Identify the area of focus.

Step 7: Facial Recognition.

- Contrast the cropped photographs with the ones from the facial database.
- Record the attendance on the attendance server.

(a)If any other face, proceed to ii) of step 6.

Step 8: End

The algorithm demonstrates the sequential operation of a system. We will require the subsequent algorithm for this system.

IV. RESULT AND DISCUSSION

In earlier days monitoring system was tough to detect the position of student and it is highly laborious work to take attendance manually. As technology has improved, integrating the monitoring system with an automated technology would provides more comfortable manner in monitoring the student. The Radio Frequency Identification (RFID) technology is one of an automated technology that is effective in upgrading existing traditional way of monitoring pupil.

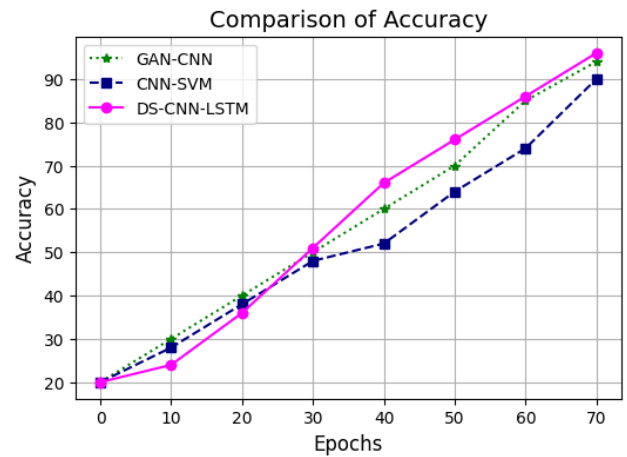


Fig. 4. Comparison of Accuracy

Figure 4 presents a comparison of the proposed system with prior state-of-the-art facial recognition methods. The proposed algorithm demonstrably outperformed at each epoch regarding mean accuracy.

TABLE I. TRAINING RESULT

Model s	LR	EPOCH H	Validation n Accuracy	Estimation n Time	Min. Iteration n
GAN-CNN	0.0002	20	93.24	23min 0sec	240
DS-CNN-LSTM	0.0002	25	95.32	15min 32sec	280
CNN-SVM	0.0002	20	89.42	20min 12sec	240

Table 1 employed the identical initial learning rate and maximum iterations for the networks, however the number of epochs varied for each network. DS-CNN-LSTM is the optimal network for data training, since it demonstrates the highest validation accuracy, although requiring the longest training duration.

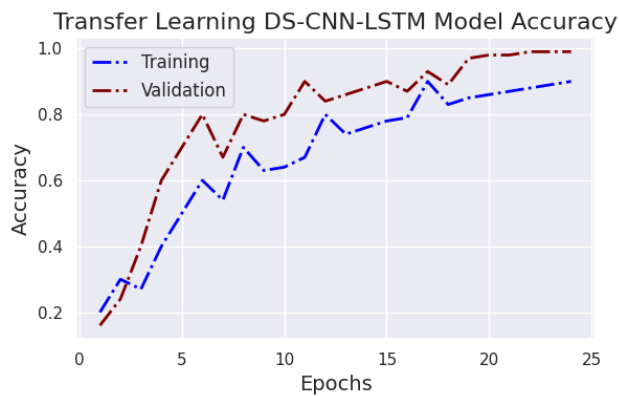


Fig. 5. TL DS-CNN_LSTM Accuracy of the Model

Figure 5 illustrates that accuracy augmented with the increase in epochs, signifying an enhancement in the model's capacity to predict class labels.

TABLE II. EVALUATION OF TRAINING AND VALIDATION

Combination Models	Recognition Rate(%) of Rank 1	
	Training	Validation
GAN-CNN	95.49	93.24
DS-CNN-LSTM	97.56	95.32
CNN-SVM	93.53	89.42

As illustrated in Table 2. The combination of DS-CNN and LSTM models yields superior results compared to the other three models, with a training accuracy of 97.56%. Moreover, hyperparameter adjustment can yield optimised values for the models.

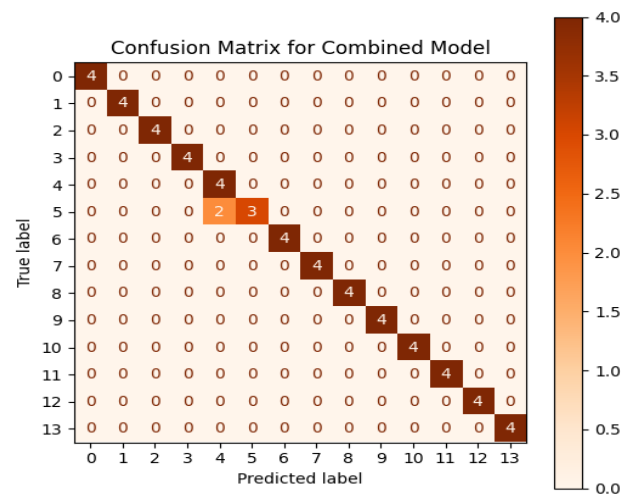


Fig. 6. Confusion Matrix

Figure 6 demonstrates that the DS-CNN-LSTM model correctly identified all instances, with the exception of class 7. For example, one occurrence that correctly belonged to class 7 was erroneously classified as class 4.

TABLE III. COMPARISON OF EVALUATION

	DS-CNN-LSTM	GAN-CNN, CNN-SVM
TA	97.56	94.50
VA	95.32	91.33
Testing Ac	96.49	91.33
TL	-	0.3522
VL	-	0.0812
Testing Loss	-	0.0756
Precision	94.50	92.61
Recall	92.17	90.21
F1-Score	96.50	94.40

Nonetheless, superior accuracy on the validation set compared to training and testing accuracy may be attributed to reduced noise in the samples resulting from the limited size of the validation set utilised. The comprehensive assessment comparison of the models is illustrated in Table 3.

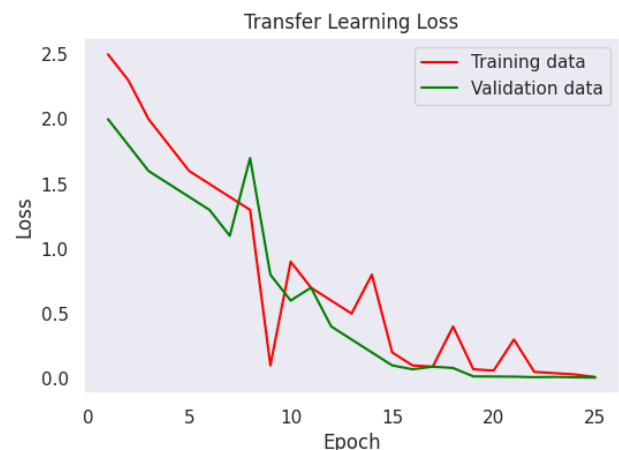


Fig. 7. Transfer Learning Loss for Model DS-CNN-LSTM

Figure 7 illustrates the model loss for both the training set and validation set across each epoch. The model underwent training for 25 epochs, although the process was halted at the 20th epoch.

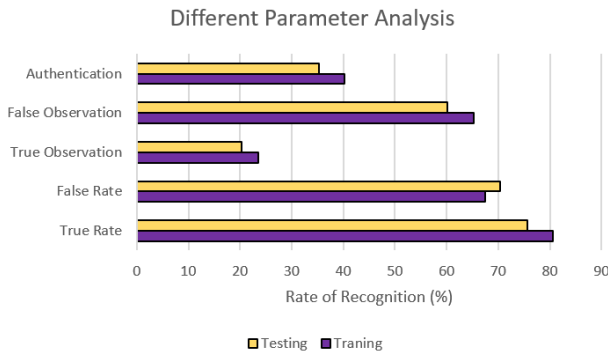


Fig. 8. Analysis for Different Parameter

The comprehensive study has been conducted at various levels of training and testing using the proposed model to attain enhanced results shown in Figure 8. The actual observation during training is 23.54%, while the observation during testing is 20.34%. The authentication rate is 40.18%, and the testing rate is 35.25%.

V. CONCLUSION

The attendance system was implemented to uphold discipline and ensure that pupils get maximum information in schools, colleges, and universities. Two traditional methods exist for recording student attendance in a certain class. One method involves calling the roll number, while the second is having pupils sign a sheet of paper according to their roll number. Through an analysis of the issues at hand, a face recognition technology-based attendance system is offered, along with research on the face recognition attendance system. The face recognition time attendance system is more stable and accurately identifies check-ins, significantly reducing the rate of class skipping compared to the control group. Feature extraction and classification processes utilising the LBPH algorithm. The system exhibits outstanding real-time efficacy in tracking numerous faces within a single frame, particularly excelling in attendance verification. The DS-CNN-LSTM model has exceptional performance, with an accuracy of 96.43% and an F-score of 96.50% when integrated with GAN-CNN and CNN-SVM.

REFERENCES

- [1] F. Allassery, "A Smart Classroom of Wireless Sensor Networks for Students Time Attendance System," *2019 9th IEEE Integr. STEM Educ. Conf. ISEC 2019*, pp. 324–331, 2019, doi: 10.1109/ISECon.2019.8881998.
- [2] A. S. Nadhan *et al.*, "Smart Attendance Monitoring Technology for Industry 4.0," *J. Nanomater.*, vol. 2022, 2022, doi: 10.1155/2022/4899768.
- [3] *et al.*, "Face Recognition based smart attendance system using IOT," *Int. J. Comput. Sci. Eng.*, vol. 6, no. 5, pp. 871–877, 2018, doi: 10.26438/ijcse/v6i5.871877.
- [4] E. Manigandan, M. S. Al Ansari, P. Nuthakki, S. Bhuvneshwari, G. Kala Priyadharshini, and M. Ponnusamy, "A Novel Method for Attendance Marking System Using Hybrid LSTM and RNN Based Networks," *7th Int. Conf. Electron. Commun. Aerosp. Technol. ICECA 2023 - Proc.*, no. Iceca, pp. 577–582, 2023, doi: 10.1109/ICECA58529.2023.10395861.
- [5] S. Sawhney, K. Kacker, S. Jain, S. N. Singh, and R. Garg, "Real-Time Smart Attendance System using Face Recognition Techniques," *2019 9th Int. Conf. Cloud Comput. Data Sci. Eng.*, pp. 522–525, 2019.
- [6] L. Saha, S. Birajdar, D. Uike, M. D. Kumar, G. Sivakumar, and U. A. Nayak, "LSSVM Analysis of Competency Effects on Supply Chain Management from Individual and Collaborative Perspectives," *2024 Int. Conf. Data Sci. Netw. Secur.*, pp. 1–6, 2024, doi: 10.1109/ICDSNS62112.2024.10691117.
- [7] I. Nayak, S. Bansal, P. Kale, S. Tonpe, C. S. Kumar, and R. Rastogi, "Application of RNN-ET-Based Digital Education Using Speech Interactions," *Int. Conf. Self Sustain. Artif. Intell. Syst. ICSSAS 2023 - Proc.*, no. Icssas, pp. 80–85, 2023, doi: 10.1109/ICSSAS57918.2023.10331628.
- [8] S. Balu Ambadas, M. Allah Khan, E. M. N. Sharmila, S. Singh, K. K. Senthilkumar, and K. Vigneshwaran, "Predicting Student Performance and its Outcomes in the Modern Education Systems Using Attention Based BiRNN and CNN Method," in *International Conference on Sustainable Communication Networks and Application, ICSCNA 2023 - Proceedings*, 2023, no. Icsna, pp. 1118–1123, doi: 10.1109/ICSCNA58489.2023.10370657.
- [9] S. Madumidha, A. Tripathi, P. Srinivasan, S. R. Wategaonkar, A. Chauhan, and D. Suganthi, "Interpreting the Evidence on Life Cycle to Improve Educational Outcomes of Students Based on Generalized ARC-GRU Approach," *2024 3rd Int. Conf. Innov. Technol. INOCON 2024*, pp. 1–6, 2024, doi: 10.1109/INOCON60754.2024.10511743.
- [10] M. Chauhan *et al.*, "Investigating Personalized Learning Paths to Address Educational Disparities Using Advanced Artificial Intelligence Systems," *Int. Conf. E-Mobility, Power Control Smart Syst. Futur. Technol. Sustain. Solut. ICEMPS 2024*, pp. 1–6, 2024, doi: 10.1109/ICEMPS60684.2024.10559339.
- [11] T. Thulasimani, L. Saha, B. Poreddy, V. K. Maharram, V. Prabakaran, and A. K. Lamba, "Advanced Analysis of Social and Psychological Factors in Higher Education Institutions using DNN and LSTM," *2024 Int. Conf. Data Sci. Netw. Secur.*, pp. 1–6, 2024, doi: 10.1109/ICDSNS62112.2024.10691260.
- [12] T. R. Rajesh, L. N. Kuchipudi, A. Dammalapati, and R. Surendran, "Attendance System Based on Facial Recognition Using OpenCV," *2nd Int. Conf. Sustain. Comput. Smart Syst. ICSCSS 2024 - Proc.*, vol. 9, no. 03, pp. 1529–1535, 2024, doi: 10.1109/ICSCSS60660.2024.10624932.
- [13] GHAZAL, Mohammed, Roaa Albasrawi, Najwan WAISI, and Mayyada AL HAMMOSHI, "Smart Meeting Attendance Checking Based on A multi-biometric Recognition System." *Przeglad Elektrotechniczny* 98, no. 3 (2022).
- [14] H. Yang and X. Han, "Face recognition attendance system based on real-time video processing," *IEEE Access*, vol. 8, pp. 159143–159150, 2020, doi: 10.1109/ACCESS.2020.3007205.
- [15] R. Hartanto and M. N. Adji, "Face recognition for attendance system detection," *Proc. 2018 10th Int. Conf. Inf. Technol. Electr. Eng. Smart Technol. Better Soc. ICITEE 2018*, pp. 376–381, 2018, doi: 10.1109/ICITEED.2018.8534942.
- [16] Z. Huang *et al.*, "DSCNN-LSTMs: A Lightweight and Efficient Model for Epilepsy Recognition," *Brain Sci.*, vol. 12, no. 12, 2022, doi: 10.3390/brainsci12121672.
- [17] P. Wagh, R. Thakare, J. Chaudhari, and S. Patil, "Attendance system based on face recognition using eigen face and PCA algorithms," *Proc. 2015 Int. Conf. Green Comput. Internet Things, ICGCIoT 2015*, pp. 303–308, 2016, doi: 10.1109/ICGCIoT.2015.7380478.